Real-Time Streaming Challenges in Internet of Video Things (IoVT)

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***Abstract*— The limited capabilities of Internet of Things (IoT) devices make real-time video streaming a major challenge. Video encoding and transmission are computationally intensive processes. Applications; like urban surveillance and health care monitoring, require real-time high definition video streams. Current video encoders are not designed to meet these requirements, thus alternative architectures, algorithms and compression techniques are needed to meet these goals. In this paper, real-time video streaming challenges for IoT applications are presented.**

Keywords—IoT; IoVT; video encoding; streaming; real-time encoding.

I. INTRODUCTION

Internet of Things (IoT) involves the interconnection and communication between different devices, sensors, and objects (Things). The goal of IoT is to provide better solutions and enhance services with streams of information from different sources. These objects might include different types of embedded sensors and actuators[1]. The design and deployment of IoT device, services and networks fall outside the realm of traditional computing environments. The computing elements are more integrated, pervasive, and ubiquitous. This integrated nature of IoT devices allows it to have a wide range of applications such as real-time monitoring, smart-home, smart-infrastructure, and health management to Cyber-security applications [1].

National research organizations and technology companies are investing in IoT research. The National Science Foundation (NSF) has introduced programs to support the research and development of IoT. NSF encourages proposals to address key challenges that cover the full range of IoT applications. Since it is expected that there will be more than 50 billion devices connected by 2020[2], the technology industry including AT&T and IBM and the shared global IoT alliance [3]; Cisco [4] [5]; and Google IoT Cloud platform [2] is participating in these efforts with substantial amount of investment directed towards IoT.

With IoT covering a wide range of applications, there are some difficulties that require research efforts and further development. Despite the wide spread of video applications, video encoding and processing are still considered a complex and computationally intense process. This makes Video integration into IoT applications a challenge [6][7][3][8]. To encode and process video streams, relatively powerful processing elements are required. Mobile and wirelessly connected devices have a much less computation capability and power resources than regular computing platforms. Meanwhile these devices are the most used category of devices for video applications. IoT and mobile devices share a lot of the same constraints for video applications, which imply that some aspect of technologies can be transferable to the IoT domain. This technology transfer can help the development of IoT devices with some projections predicting that over 50% of large businesses will implement some sort of IoT [9].

The Internet of Video things (IoVT) is the use of IoT devices with Video sensors on board. This distinction is needed and necessary since IoVT has a very different requirements than traditional IoT devices. The main challenges for IoVT devices and applications can be categorized into device level, communication level and platform level. The next sections will discuss in details the challenges and limitations of current IoT devices and review research conducted to overcome these limitations.

II. CHALLENGES AND CURRENT RESEARCH

A. Video Encoding

Video encoding focuses on reducing redundancies in spatial and temporal correlations. Spatial compression is performed in the Intra frame compression stage, while temporal compression is done in inter frame compression stages. The Inter frame stage uses multiple frames with motion estimation algorithms, to reduce temporal redundancies. These are known to be memory and computationally intensive processes but do provide higher compression ratios. On the other hand, Intra coding focuses on reducing the spatial redundancies in a single frame while not affect neighboring frames. Almost all of encoding standards use predictive coding in both inter and intra compression stages [6] [10] [11].

H.264 and its successor HEVC are the most used video compression standards in the market today. A study in [12] was performed to analyze the performance differences between the two standards. HEVC, which is known to be 1.2 -

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3.2 more complex that H.264, comes with the advantage of higher compression and encoding efficiency by reducing bit-rate up to 40% on average [13]. The reduced bit-rate means less energy needed to transmit video streams, but this reduction is achieved by the increase in complexity. In [12] a comparative study between HEVC and H.264 by comparing different Bjøntegaard-Delta Rate (BD-Rate) results of different video test sequences from different classes. The study shows that HEVC consumes 17.4% more energy on average while reducing bit-rate by 25%. The authors conclude that for most applications HEVC is more efficient that H.264. Although H.264 might be more suitable for IoVT, HEVC does show far better performance but requires improvements to reduce its complexity.

In [13], an analysis of HEVC for energy reduction techniques for real-time encoders was performed. The Two main computation complexity sources for HEVC are the use of Coding Tree Units (CTU) quad tree partitioning instead of the traditional coding blocks as in H.264. Also, HEVC Intra-prediction modes are increased to include 33 angular modes in addition to the planar and DC modes with a total of 35 modes compared to 9 modes in H.264. In [13], experiments show that at different granularity levels the maximum improvement to the Intra-prediction process will yield 30% while maximum improvement to the CTU partitioning can reach up to 78.1% which can point to the amount of complexity in that process.

Real time video streaming process contains a lot of interconnected elements that can be investigated for improvement. As stated in [11][14][6] current video compression codecs mostly use Discrete Cosine Transform (DCT) as the main transformation technique. DCT is known to be a block-based transformations which has certain quality issues such blocking artifacts [14][6]. While other transformations such as Discrete Wavelet Transformation (DWT) is known to outperform DCT in Signal to Noise Ratios (SNR) and provide inherent scalability [14] [6] [10] [11]. An Intra encoding enchantment technique can be shown in [11]. The proposed Hybrid Wavelet-DCT algorithm adds wavelet-based transformation into the Intra-frame prediction process in H.264 encoding process. This adds the advantage of wavelet compression properties to create a video transmission that guarantees lower bit rates while maintaining the quality and Peak Signal to Noise Ratio (PSNR). Although this algorithm maintained a comparable PSNR values, it introduces an increase level of complexity that can hinder its usage in IoVT streaming.

IoVT devices will require improvement to the video encoding process in general; especially to accommodate connection unreliability and errors. One of the suggested modifications as mentioned in [15, 2] is to use slices and redundant slices in encoding. A slice is a group of blocks grouped and encoded together independent from other slices. The Redundant slices can be encoded in lower quality to provide a level of redundancy in case of error and packet loss. This method offloads the complexity of video encoding to the decoding side and can be applied offline [16].

IoVT real-time Streaming using resources constrained hardware is little explored. In the next section, literature review shows the technical challenges and proposed solutions to IoVT real-time streaming.

B. IoT Video streaming experiements

In [9], open source hardware, NoSQL database, Cloud and IoT platform were used. The platform used MongoDB for document storage and capture images and transferred video to a remote server which acted as the cloud. Arduino Yún with OpenWRT-Yun operating system was used. OpenWRT is a Linux based operating system which is designed to work with embedded devices [9]. The experiment succeeded in stream video wirelessly to a web application. The camera resolution used in this experiment is 640X480 pixels which is considered low for use in many application.

Another experiment is shown in [17], where the feasibility of IoT video streaming is measured in respect to the delay and overhead. The experiment uses Peer-to-Peer (P2P) network and “sensibleThings” platform for IoT network layer. H.264 was used for the video encoding process. H.264 is an attractive option due to the availability of H.264 encoding and decoding units on the raspberry pi platform which was used in this experiment. The experiment used (P2P) with Distributed Hash Tables (DHT); according to [17] this approach allowed to send minimal chunks of encoded videos as small as P2P packets to minimize delay of video transmission. But even with these enhancements, the video encoding processes consumed more than 90% of the total delay.

Multiple solutions were investigated to decrease the transfer time and possibly off-loading to a cloud based service. In [18], a fog computing platform was used with urban traffic surveillance as a case study. A dynamic video stream processing scheme is proposed to meet the requirements of real-time information processing. Fog computing was proposed as a solution to tackle the transfer delay problem that is associated with using cloud computing and to provide more processing power to the IoT devices. Wide Area Motion Imagery (WAMI) was used which is characterized by its high data rates and wide area coverage[18]. This is an important requirement for any real-time urban surveillance application. A key point to note is the size of the frame that is produced by such detailed camera. The experiment used techniques in which only a sub-area that is tracked in the frame is sent back to the processing station not the whole frame. The sub-area is decided on the fog unit which provides a more computationally capable unit. This edge-based architecture allows for video streams to be processed on the edge to minimize transmission and only send metadata to control units.

Another approach is to use current established systems to accelerate the deployment of IoT services. One example can be seen in [19], which draws from existing image processing pipelines and modules from mobile devices. The authors discuss the smartphone image processing flow, where much of the raw image processing devolved into an image processing pipeline (IPP). The IPP extracts and pre-processes image frames prior to compression. Preprocessing includes de-mosaicking and multi-frame operations. Often these operations are done on a dedicated Image Signal Processing unit (ISP) [19]. All these functionality are done independent of the main processing unit of the IoT device.

Although these experiments show a possibility for streaming for IoT, delay and computation complexity are still main problems for real-time streaming for IoVT. The following section discusses a platform proposal for IoVT with suggested improvements and modifications.

I. PROPOSED PLATFORM DEVELOPMENT FOR IOVT

IoVT device constraints and limitations require improvement in multiple blocks to facilitate real-time streaming. There are different approaches and inter-disciplinary fields to consider, including distributed/partial video encoding, transform and compression efficiency, Intra-prediction process, and video streaming over low power communication technologies, e.g. Zig-Bee, sensor networks.

A. *Encoding and Real-time streaming*

The most common used video encoding standard is H.264. While H.264 standard is widely implemented, HEVC standard is more efficient and reduces 40% of the bit on average compared to H.264. But HEVC is more complex and more computationally intensive than the H.264 [20] [21]. The bit-rate reduction is desired for IoVT applications. With further improvement in HEVC especially in the Intra-prediction and CTU partitioning, HEVC can be a suitable standard for IoVT.

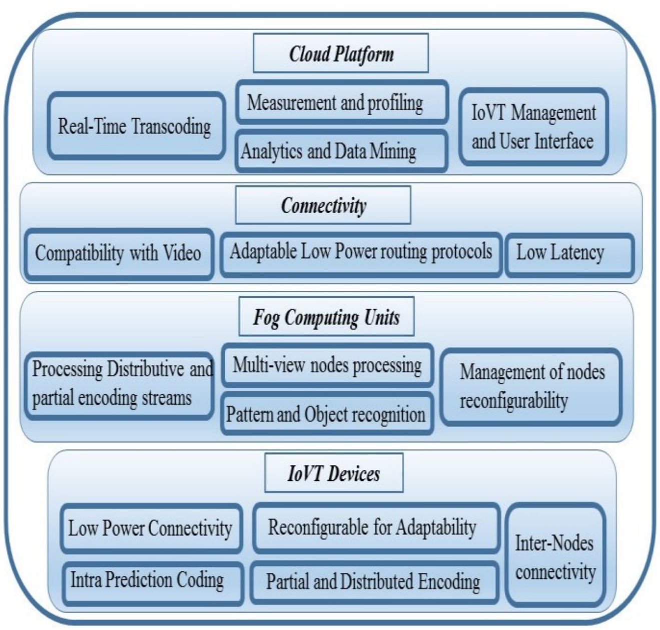

Fig 1: Overview of IoVT platform elements

Current investigations include a study of using different transforms coupled with current known standards to see if a simpler computationally flexible encoder is possible. The initial investigation uses DWT for transforms, which is attractive because it has different attributes that can be used with different applications, examples can be found at [18]. A multi-dimensional (3-D) wavelet transformation is used to encode video sequence using swarm intelligence. The idea is to apply 3D DWT to the input video sequence, and by using Particle Swarm Optimizations (PSO) algorithms to choose the most optimal sub-bands to be kept and encoded. Other examples can be seen in the use of wavelet transformation in video sequences to detect fire using color analysis [22]. Also wavelet can be used to provide better encryption and security for video feeds by securing video blocks instead of video stream. This is important since security is another issue facing IoT streaming and it shows potential hardware saving and more efficient implementations [23].

B. *IoVT Device*

When looking at the two most common devices used for IoT prototyping, Arduino Yún and Raspberry Pi are most common choices. Arduino Yún is equipped with Atheros AR9331 and 64 MB of RAM [24], while Raspberry Pi 3 model B comes with Broadcom BCM2837 chipset and 1GB SDRAM [25]. These specs are to be considered the high-end for IoT devices. Other approach include offload some of the video coding and compression computations to the distributed-nodes or to a fog computing unit. This is an attractive feature for IoVT platform especially with higher resolution and lower latency requirements. If Intra-Prediction compression is done on the device side, conversion, transcoding and additional processing are feasible on fog computing units while heavier processing is done on a cloud platform [3][8][12].

C. *Connectivity*

Connection using low power methods impacts the streaming efficiency, protocols needs to take into consideration video streaming format and error recovery. The traditional Internet connections are not suited for video streaming due to TCP congestion control with data flow and rate of transmission restrictions. Also, there is no support for end-to-end video streaming Quality of Service (QoS) [15][3][8][26] which is a needed quality for IoVT applications. Connection layer should adhere to the general rules of real-time streaming as in [28], such as: Keep the Data moving, Handle stream Imperfections, Generate predictable outcomes, and Guarantee data safety and availability. Another major important factor is securing connections of IoVT nodes, since these are video feeds and are inherently sensitive for most applications. As mentioned earlier, connection encryption can be embedded into the video encoding process to eliminate any additional computation workload.

D. *Fog and Cloud Computing elements*

Cloud and fog computing platforms are utilized to perform the heavy computations such as analytics, object and pattern recognition, profiling and measurements of the quality of the video stream. Cloud platform combined with the re-configurability of devices can make predictive decisions to optimize and make the necessary changes to enhance performance and lower power consumption. The use of reconfigurable computing is becoming popular with IoT devices with companies like Xilinx and Altera leading the way [26][27]. Reconfigurable IoT devices will be able to handle changing performance requirements and adapt to the changing demands such as connection status and power resources.

II. Conclusion

In this paper, IoVT real-time video streaming challenges are presented. Different literature experiments and show-cases highlights the limitation of the current IoT video streaming implementations. A platform for real-time streaming IoVT is proposed with emphasis on elements that effects the reliability and efficiency of video streaming.

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